

MS&E 228: Applied Causal Inference Powered by ML and AI

Lecture 7: Variance of Linear Adjustment and Interactive OLS in RCTs; Initiating study of Continuous Treatments

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Readings: *Applied Causal Inference Powered by ML and AI*, §1-2.

Four Key Takeaways for Today

1. **OLS coefficient = ATE in RCTs:** When using $\text{OLS}(Y \sim D, X)$ in an RCT, the coefficient on D converges to the ATE even if the CEF is not linear—this follows from the equivalence of OLS and DR in this setting.
2. **Simple OLS can hurt precision:** Adjusting with $\text{OLS}(Y \sim D, X)$ can *increase* variance compared to the two-means estimator when the experiment is unbalanced ($q \neq 1/2$).
3. **Interactive OLS guarantees improvement:** Running $\text{OLS}(Y \sim D, D \cdot X, X)$ ensures variance reduction over two-means for *any* treatment probability q .
4. **Continuous Treatments:** We need to make some modeling assumptions, as otherwise the problem is statistically daunting and the overlap assumption is quite restrictive.

Part I: DR with Simple OLS in RCTs

Connecting OLS to Doubly Robust Estimation

Where We Left Off

- ▶ Last lecture, we saw that the DR estimator in an RCT recovers the ATE even if the outcome regression $\hat{g}$ converges to a limit $g_* \neq g_0$ (the true CEF).
- ▶ This led to a key open question:

Key Question

Is the variance of the DR estimator with a misspecified g_* better or worse than the simple two-means estimator?

Today, we analyze this question for a very common choice: when $\hat{g}$ is estimated via simple OLS.

Setup: Simple Linear Regression for g

We run OLS of Y on D and X to get estimated coefficients $\hat{\theta}, \hat{\beta}, \hat{c}$ that minimize the empirical squared loss:

$$\min_{\theta, \beta, c} \mathbb{E}_n[(Y - \theta D - \beta' X - c)^2]$$

This is equivalent to satisfying the empirical **normal equations**:

$$\mathbb{E}_n[\underbrace{(Y - \hat{\theta}D - \hat{\beta}'X - \hat{c})}_{\hat{\epsilon}} \cdot (1; D; X)] = 0$$

Let $\hat{\epsilon}$ be the empirical OLS residuals. The normal equations state that these residuals are empirically orthogonal to the regressors.

OLS Coefficient = DR Estimator in RCTs

- **G-formula with linear OLS:** The ATE estimate is simply the coefficient on D:

$$\hat{\theta}_g = \hat{\theta}$$

- **Doubly Robust formula:** We plug in $\hat{g}(D, X) = \hat{\theta}D + \hat{\beta}'X + \hat{c}$:

$$\hat{\theta}_{\text{DR}} = \mathbb{E}_n \left[\hat{g}(1, X) - \hat{g}(0, X) + \frac{D - q}{q(1 - q)} (Y - \hat{g}(D, X)) \right] = \mathbb{E}_n \left[\hat{\theta} + \frac{D - q}{q(1 - q)} \cdot \hat{c} \right]$$

Key Insight

The correction term vanishes! The DR estimator is numerically identical to the OLS coefficient $\hat{\theta}$.

$$\mathbb{E}_n \left[\frac{D - q}{q(1 - q)} \hat{c} \right] = \frac{1}{q(1 - q)} (\mathbb{E}_n[D\hat{c}] - q\mathbb{E}_n[\hat{c}]) = 0$$

This is a direct consequence of the normal equations: $\mathbb{E}_n[D\hat{c}] = 0$ and $\mathbb{E}_n[\hat{c}] = 0$.

A Non-Trivial Theorem

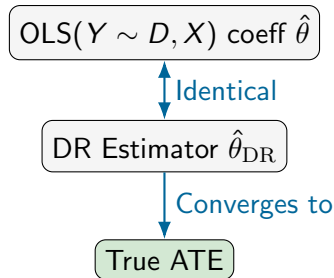
This equivalence leads to a powerful and practical result:

Theorem

The coefficient $\hat{\theta}$ on D in an OLS regression of $Y \sim D, X$ converges to the true ATE when applied to RCT data, **even when the true Conditional Expectation Function (CEF) of Y given D and X is not linear.**

Intuition:

- ▶ We know the DR estimator converges to the ATE in an RCT (if $\hat{g}$ converges to *some* limit).
- ▶ We just showed the DR estimator is identical to the OLS coefficient $\hat{\theta}$.
- ▶ Therefore, $\hat{\theta}$ must also converge to the ATE.



Part II: When Does OLS Adjustment Improve Precision?

A Cautionary Tale

Setting Up the Variance Analysis

To analyze the variance, we look at the population analogues.

Population Best Linear Predictor:

$$g_*(D, X) = \theta D + \beta' X + c$$

This is the linear model that minimizes the population prediction loss:

$$\min_{\theta, \beta, c} \mathbb{E}[(Y - \theta D - \beta' X - c)^2]$$

Population Normal Equations: Equivalently, satisfies normal equations

$$\mathbb{E}[\underbrace{(Y - \theta D - \beta' X - c)}_{\epsilon} \cdot (1; D; X)] = 0$$

Let ϵ be the **population residuals**. From equivalent with DR, we know that $\theta = \text{ATE}$.

Variance of the Linearly Adjusted Estimator

Since the DR estimator is equivalent to the OLS coefficient, we can use the DR variance formula. The asymptotic variance of $\sqrt{n}(\hat{\theta} - \theta)$ is:

$$V_{\text{OLS}} = \text{Var} \left(\theta + \frac{D - q}{q(1 - q)} \cdot \epsilon \right)$$

Because $\mathbb{E}[\epsilon] = 0$ and $\mathbb{E}[\epsilon \cdot D] = 0$ from the population normal equations:

$$V_{\text{OLS}} = \frac{\mathbb{E}[(D - q)^2 \epsilon^2]}{(q(1 - q))^2}$$

Connection to Robust Standard Errors

This is exactly the formula for **heteroskedasticity-robust (HC0)** standard errors for the parameter θ . It is a simplified version of the famous "**sandwich formula**".

Python Pseudocode: Simple OLS Adjustment

```
import numpy as np
import statsmodels.api as sm

# 1. Fit the OLS model
X_reg = sm.add_constant(np.c_[D, X])
model = sm.OLS(Y, X_reg).fit()

# 2. The ATE estimate is the coefficient on the treatment D
# Note: Index is 1 because column 0 is the constant
ate_estimate = model.params[1]

# 3. Get heteroskedasticity-robust standard errors
# This is an estimate of the DR variance formula
ate_se = model.get_robustcov_results(cov_type='HCO').bse[1]

print(f"ATE Estimate: {ate_estimate:.4f}")
print(f"Robust S.E.: {ate_se:.4f}")
```

Key Question

How does this variance compare to the asymptotic variance of the two-means estimator?

Variance of the Two-Means Estimator

The variance of the simple two-means estimator ($\bar{Y}_1 - \bar{Y}_0$) is:

$$\text{Var}(\text{Two-Means}) = \frac{\text{Var}(Y|D=1)}{q} + \frac{\text{Var}(Y|D=0)}{1-q}$$

Key Insight

Two-Means is equivalent to OLS($Y \sim D$), which learns a model $\hat{\theta}D + \hat{\theta}_0$. $\hat{\theta}$ converges to ATE and $\hat{\theta}_0$ to $\mathbb{E}[Y | D=0] = \mathbb{E}[Y(0)]$.

Apply, equivalence of Two-Means to DR with outcome regression $g_*(D, X) = \theta D + \theta_0$, to derive variance formula (verify that $\text{Var}(\text{Two-Means}) = V_{\text{TM}}$ as an exercise):

$$V_{\text{TM}} = \frac{\mathbb{E}[(Y - \theta D - \theta_0)^2(D - q)^2]}{(q(1 - q))^2}$$

Key Observation: Formula has same structure as V_{OLS} , but the residuals are different.

Comparing the Variances

Let's relate the two residuals. Let $\epsilon_{\text{TM}} = Y - \theta D - \theta_0$. Then:

$$\epsilon_{\text{TM}} = (Y - \theta D - \beta'X - c) + (\beta'X + c - \theta_0) = \epsilon + (\beta'X + c - \theta_0)$$

Plugging this into the variance formula for the two-means estimator:

$$V_{\text{TM}} = \underbrace{\frac{\mathbb{E}[\epsilon^2(D - q)^2]}{(q(1 - q))^2}}_{V_{\text{OLS}}} + \underbrace{\frac{\mathbb{E}[(\beta'X + c - \theta_0)^2(D - q)^2]}{(q(1 - q))^2}}_{\text{Positive Term}} + 2 \underbrace{\frac{\mathbb{E}[\epsilon(\beta'X + c - \theta_0)(D - q)^2]}{(q(1 - q))^2}}_{\text{Problematic Cross-Term}}$$

If the cross-term were zero, we would be done! V_{DR} would always be smaller.

The Problematic Cross-Term

The cross-term contains expressions like:

$$\beta' \mathbb{E}[\epsilon \cdot X \cdot D]$$

From the population normal equations for OLS($Y \sim D, X$), we know:

- ▶ $\mathbb{E}[\epsilon \cdot D] = 0$ ✓
- ▶ $\mathbb{E}[\epsilon \cdot X] = 0$ ✓
- ▶ $\mathbb{E}[\epsilon] = 0$ ✓

But...

We **do not** know that $\mathbb{E}[\epsilon \cdot X \cdot D] = 0$! This would only be true if the CEF were truly linear (i.e., $\mathbb{E}[\epsilon|D, X] = 0$). We only have the weaker "orthogonality" conditions that the OLS residuals satisfy.

When OLS Adjustment Can Hurt Precision

Because $\mathbb{E}[\epsilon \cdot X \cdot D]$ is not guaranteed to be zero, the cross-term can be substantially negative.

This negative term can be large enough to overwhelm the positive term in the variance decomposition.

The Surprising Result

It is possible for $\text{Var}(\text{Two-Means}) < \text{Var}(\text{OLS Adjusted})$! Simple OLS adjustment is **not** guaranteed to improve precision.

Homework Preview: You will construct a simple data generating process where this exact scenario occurs. You will also prove that when $q = 1/2$ the problem vanishes and OLS is always weakly better than TM.

Key Question

How can we guarantee that our linear regression residual ϵ is orthogonal to $X \cdot D$, i.e., $\mathbb{E}[\epsilon \cdot X \cdot D] = 0$?

Part III: Interactive OLS for Guaranteed Improvement

The Right Way to Adjust in Experiments

Diagnosing the Problem, Finding the Solution

The Problem: Our OLS residuals did not satisfy orthogonality with respect to the product term $D \cdot X$. That is, $\mathbb{E}[\epsilon \cdot D \cdot X] \neq 0$.

The Solution: Force this orthogonality by adding the interaction term to the regression!

Recommended Specification

For analyzing RCTs, it is advisable that you run OLS with interactions:

$$\text{OLS}(Y \sim D, X, D \cdot (X - \bar{X}))$$

where $\bar{X} = \mathbb{E}_n[X]$. This is sometimes called a "fully interacted model". De-meaning the covariates is not so important but simplifies variance calculations.

The Interactive OLS Specification

This estimates a model:

$$\hat{g}(D, X) = \hat{a}_0 D + \hat{a}'_1 (X - \bar{X}) \cdot D + \hat{\beta}' X + \hat{c}$$

CATE Estimate: The Conditional Average Treatment Effect is now a function of X :

$$\widehat{\text{CATE}}(X) = \hat{g}(1, X) - \hat{g}(0, X) = \hat{a}_0 + \hat{a}'_1 (X - \bar{X})$$

ATE Estimate: We can estimate the ATE via the empirical g-formula:

$$\hat{\theta} = \mathbb{E}_n[\widehat{\text{CATE}}(X)] = \hat{a}_0 + \hat{a}'_1 (\mathbb{E}_n[X] - \bar{X}) = \hat{a}_0$$

The new OLS residuals $\hat{e}$ will now satisfy the empirical normal equations:

$$\mathbb{E}_n[\hat{e} \cdot (1; D; X; D \cdot X)] = 0$$

This guarantees $\mathbb{E}_n[\hat{e} \cdot D \cdot X] = 0$.

Variance of the Interactive OLS Estimator

Since the debiasing term in the DR formula is still zero due to normal equations, we can analyze the variance of the DR score with the new limit predictor g_* :

$$\text{Var}(a_0 + a'_1(X - \mathbb{E}[X])) + \frac{\mathbb{E}[\epsilon^2(D - q)^2]}{(q(1 - q))^2}$$

- ▶ The first term, $\text{Var}(a_0 + a'_1(X - \mathbb{E}[X]))$, is the variance of the best linear approximation of the CATE.
- ▶ The second term is the same form as before, but now with residuals ϵ that satisfy $\mathbb{E}[\epsilon \cdot D \cdot X] = 0$. This ensures this term is smaller than or equal to the variance of the two-means estimator.

Main Precision Improving Theorem

Guaranteed Precision Improvement

The overall asymptotic variance of the interactive OLS estimator is guaranteed to be less than or equal to the variance of the two-means estimator (Lin, 2013, Negi & Wooldridge, 2021).

Practical Note: How to estimate the asymptotic variance in practice?

$$\underbrace{\text{Var}(a_0 + a'_1(X - \mathbb{E}[X]))}_{V_1} + \underbrace{\frac{\mathbb{E}[\epsilon^2(D - q)^2]}{(q(1 - q))^2}}_{V_2}$$

- ▶ The second term (V_2) in the variance is exactly what the HC0 variance $\hat{V}_2$ for $\hat{a}_0$ in OLS approximates, **if you de-mean the covariates X** before passing to OLS.
- ▶ Most statistical packages can compute this directly.
- ▶ Add to this variance the empirical variance of the estimated CATE model ($\hat{V}_1$).
- ▶ Use this variance when calculating the standard error (i.e. $\hat{se} = \sqrt{(\hat{V}_1 + \hat{V}_2)/n}$).

Standard Errors for Interactive Regression: Python Code

```
# De-mean covariates and create interaction terms
X_dm = X - X.mean(axis=0)
X_interact = patsy.dmatrix('D * (' + '+'.join(X_dm.columns) + ')',
                           X_dm.assign(D=D), return_type='dataframe')

# Run OLS with HCO standard errors
ira = sm.OLS(Y, X_interact)
ira_results = ira.fit(cov_type='HCO')
ira_est = ira_results.params['D']

# Account for error in means of X (first-order effect on ATE)
interaction_cols = [c for c in X_interact.columns if c.startswith('D:')]
cate = X_dm @ ira_results.params[interaction_cols]
ira_se = np.sqrt(ira_results.HCO_se['D']**2 + np.var(cate)/len(Y))
```


Poll

Poll Question

You are analyzing an RCT with an unbalanced design (e.g., treatment probability $q = 0.3$). Which OLS specification is **guaranteed** to have variance that is at most the variance of the simple two-means estimator?

- A. $\text{OLS}(Y \sim D)$
- B. $\text{OLS}(Y \sim D, X)$
- C. $\text{OLS}(Y \sim D, X, D \cdot X)$
- D. Both A and C
- E. Both B and C



(Poll Everywhere)

Extension: ML-Learned Features (Guo et al., 2021)

We can go further by using ML-learned features $\hat{\phi}(X)$:

1. Train ML models (trees, neural nets) to predict Y from X (using cross-fitting).
2. Use their out-of-fold predictions $\Phi = \hat{\phi}(X)$ as additional features.
3. Run $\text{OLS}(Y \sim D + X + D \cdot (X - \bar{X}) + \Phi + D \cdot (\Phi - \bar{\Phi}))$.

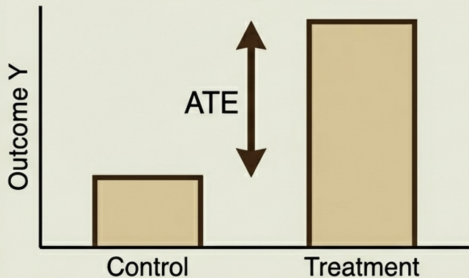
The ATE estimate is the coefficient a_0 associated with D . The variance can be calculated as in previous Interactive Regression slide.

This is used at Meta for variance reduction in A/B testing!

Part IV: Observational Data and Continuous Treatments

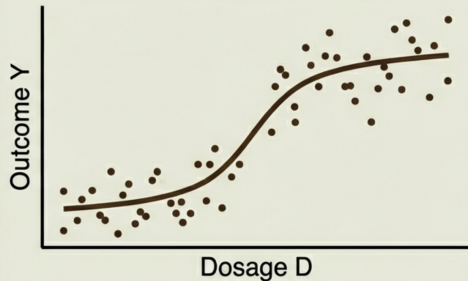
From Binary Effects to Dose-Response Curves

The Binary World



Switching (Group A vs B)

The Continuous World



Tuning (Dose, Price, Spend)

Motivation for Continuous Treatments

Many real-world questions involve treatments that are not binary.

Healthcare: What is the **dose-response-curve**? How does patient response change with the dose of a drug?

Economics: What is the **demand-curve**? How does demand for a product change with its price?

Operations Management: What is the **return on investment**? If we invest x dollars, what is our return per dollar?

Framework for Continuous Treatments

The potential outcomes framework extends naturally.

- ▶ **Potential Outcomes:** $Y(d)$, where the treatment level d is now a continuous variable. We have infinitely many potential outcomes for each individual.
- ▶ **Conditional Exogeneity:** The assumption is the same. For any level d :

$$Y(d) \perp\!\!\!\perp D \mid X$$

- ▶ **Overlap:** This assumption becomes much stronger. We need the density of the joint distribution of (D, X) to be non-zero at points (d, X) for any value d and for any X in the support. This is harder to satisfy.

Causal Estimands for Continuous Treatments

Our questions of interest also become richer.

- ▶ **Mean Counterfactual / Dose-Response Curve:**

$$\mu(d) = \mathbb{E}[Y(d)]$$

This is a curve representing the average outcome if we intervene and fix the value of the treatment at some treatment level d .

- ▶ **Marginal Effect at a point d :**

$$\text{ME}(d) = \mathbb{E} \left[\frac{\partial}{\partial d'} Y(d') \Big|_{d'=d} \right]$$

How much does the outcome change on average if we perturb the treatment around some level d .

- ▶ **Average Marginal Effect (AME):** The average of the marginal effects, taken over the *natural distribution of observed treatment values* D .

$$\text{AME} = \mathbb{E}[\text{ME}(D)]$$

Discussion Question

Group Discussion

Suppose I give you an observational dataset. I tell you that a set of features X are the main drivers of the drug dosage D (a continuous variable) that patients receive. I want you to estimate the effect of the drug dosage on the outcome Y .

What would you do? What are the first steps you would take?

Identification: G-Formula and Its Challenges

Identification via G-Formula: Under conditional exogeneity and overlap, we can still use the g-formula:

$$\mathbb{E}[Y(d)] = \mathbb{E}[\mathbb{E}[Y(d)|X]] = \mathbb{E}[\mathbb{E}[Y|D = d, X]] = \mathbb{E}[g(d, X)]$$

where $g(d, X) = \mathbb{E}[Y|D = d, X]$.

Challenge: IPW Fails

IPW-style estimators do not work in the same way. For a continuous treatment D , what is $\Pr(D = d|X)$? It's a conditional density, $f_{D|X}(d|X)$. The IPW weight would be $1\{D = d\}/f_{D|X}(d|X)$, which is zero almost everywhere. We need a different approach.

A Simple Approach: The Linear Model

Assumption. Suppose the conditional expectation of the potential outcome is linear:

$$\mathbb{E}[Y(d)|X] = \theta \cdot d + \beta'X + c$$

Under this assumption, our causal estimands simplify dramatically:

- ▶ **Dose-Response:** $\mathbb{E}[Y(d)] = \theta \cdot d + \beta'\mathbb{E}[X] + c$
- ▶ **ATE(d):** $\mathbb{E}[Y(d) - Y(0)] = \theta \cdot d$
- ▶ **Marginal Effect:** $\frac{\partial}{\partial d}\mathbb{E}[Y(d)] = \theta$ (constant)

Key Insight

All causal estimands boil down to estimating the single parameter θ .

Identification without Overlap under Linearity

For any point (d, x) in the support of the distribution, by Conditional Exogeneity:

$$\mathbb{E}[Y(d) \mid X = x] = \mathbb{E}[Y(d) \mid D = d, X = x] = \mathbb{E}[Y \mid D = d, X = x]$$

Thus by our assumption on the form of the mean counterfactual outcome:

$$\mathbb{E}[Y \mid D = d, X = x] = \theta \cdot d + \beta'x + c \quad (1)$$

Identification Strategy

If the best linear predictor of Y from D, X , has a unique coefficient associated with D , then the causal estimands can be uniquely identified without the overlap assumption.

Estimation Strategy

Simply run $\text{OLS}(Y \sim D, X)$.

Key Questions

1. When does the population best linear predictor have a unique solution with respect to θ ?
2. How can we better interpret the coefficient θ in the best linear predictor?
3. What are the drives of high-variance for the estimated parameter $\hat{\theta}$?
4. What is the analogue of “overlap” under linearity?

Summary: Four Key Takeaways

1. **OLS coefficient = ATE in RCTs:** The coefficient on D in $OLS(Y \sim D, X)$ is a consistent estimator for the ATE in an RCT, even if the CEF is not linear.
2. **Simple OLS can hurt precision:** For unbalanced experiments ($q \neq 1/2$), simple OLS adjustment can increase variance compared to the two-means estimator.
3. **Interactive OLS guarantees improvement:** The fully interacted model, $OLS(Y \sim D, X, D \cdot X)$, is guaranteed to (weakly) reduce variance.
4. **Continuous Treatments:** We need to make some modeling assumptions, as otherwise the problem is statistically daunting and the overlap assumption is quite restrictive.

References

- ▶ Lin, W. (2013). Agnostic notes on regression adjustments to experimental data: Reexamining Freedman's critique. *The Annals of Applied Statistics*, 7(1), 295-318.
- ▶ Negi, A., & Wooldridge, J. M. (2021). Revisiting regression adjustment in experiments with heterogeneous treatment effects. *Econometric Reviews*, 40(5), 463-485.
- ▶ Guo, H., et al. (2021). Machine Learning for Variance Reduction in Online Experiments. *arXiv preprint arXiv:2106.07263*.